

Shadow length

W21 (2021) — English translation

As part of this task, you will have to investigate how the length of the shadow of a light black ball tied to the bottom of a wide vessel with a thread and illuminated by sunlight depends on the level of transparent liquid h in the vessel. The length of a shadow is defined as the maximum distance between its boundary points. Consider that the Archimedes force acting on the ball in the air is sufficient to keep the thread taut.

In what follows, use the following notation:

1. Length of the thread l
2. Radius of the ball R
3. Angle of incidence of the rays on the surface of the liquid α
4. Angle of refraction of the rays β
5. Length of the shadow L

The graph shows the dependence of the shadow length on the liquid level in the vessel.

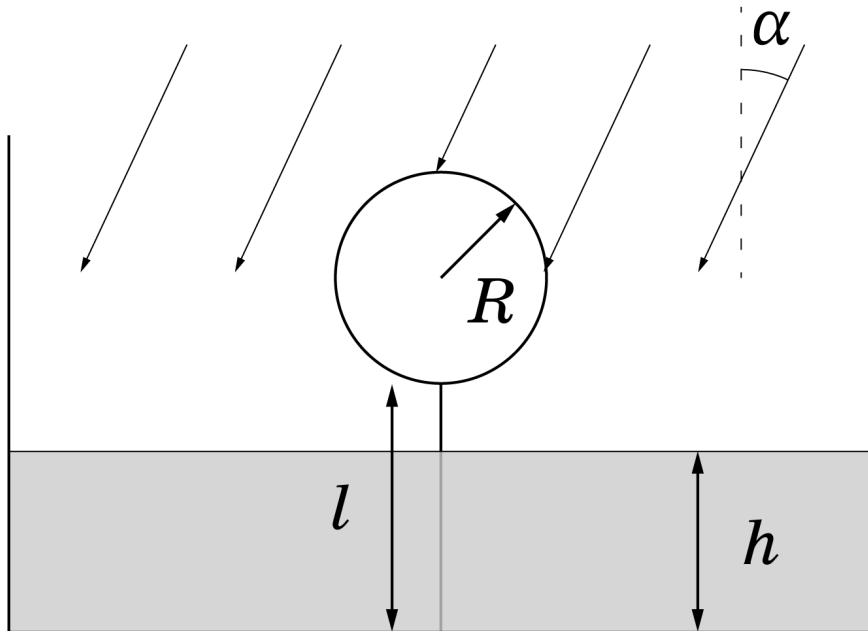


Figure 1: Figure 1.

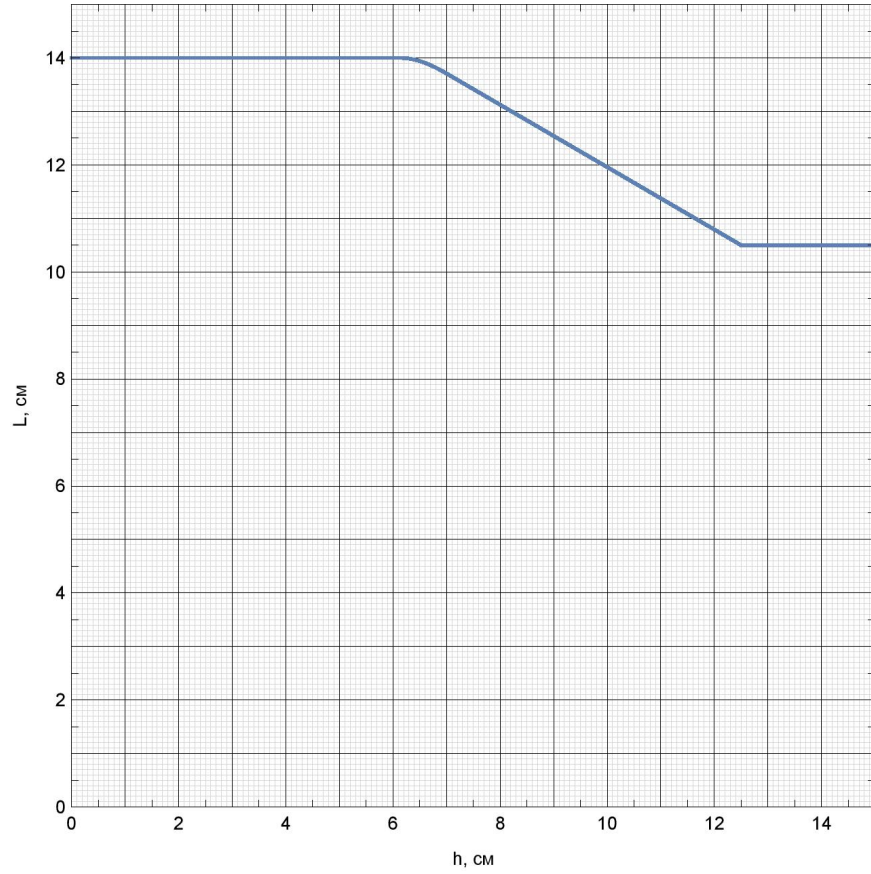


Figure 2: Figure 2.

Part A. Before the Break

A1 Express L in terms of R and α with $h = 0$.

0.20

Let h_1 be the water level in the vessel at the moment when the break begins in the graph.

A2 Express L in terms of R , α , β , l and h with $0 < h < h_1$.

0.30

A3 Express h_1 in terms of l , R , α and β .

1.00

Part B: Fracture Study

Let h_2 be the water level at the moment when the kink turns into the linear section of the dependence.

B1 For $h_1 < h < h_2$, obtain the dependence of the shadow length on l , R , α , β and h .

2.50

B2 Express h_2 in terms of l , R , α , β .

1.00

Part C. Linear Section

Let h_3 be the liquid level at the moment when the length of the shadow stops changing.

C1	Since the section is linear, the dependence $L(h)$ has the form $L = L_0 - kh$. Express L_0 and k in terms of l , R , α and β .	1.30
C2	Express the minimum shadow length in terms of l , R , α and β .	0.20
C3	Express h_3 in terms of l , R , α and β .	0.50

Part D. Numerical values

D1	Find the numerical values of $\cos \alpha$ and $\cos \beta$.	1.50
D2	Find the radius of the ball R .	0.50
D3	Find the refractive index of the liquid n .	0.50
D4	Find the length of the thread l .	0.50

Source: <https://pho.rs/p/70>